# Decision Theory

BY

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# Which Attribute is ”best”?

- We would like to select the attribute that is most useful for classifying examples.
- • ***Information gain*** measures how well a given attribute separates the training examples according to their target classification.
- • ID3 uses this *information gain* measure to select among the candidate attributes at each step while growing the tree.
- • In order to define information gain precisely, we use a measure commonly used in information theory, called ***entropy***
- • ***Entropy*** characterizes the (im)purity of an arbitrary collection of examples.

# Information Theory –ID3 (Iterative Dichotomiser 3)

- ID3 algorithm invented by Ross Quinlan and uses information gain as its attribute selection measure
- This measure is based on pioneering work by Claude Shannon on information theory, which studied the value or "information content" of messages
- Let node N represent or hold the tuples of partition D. The attribute with the highest information gain is chosen as the splitting attribute for node N
- This attribute minimizes the information needed to classify the tuples in the resulting partitions and reflects the least randomness or "impurity" in these partitions
- The expected information needed to classify a tuple in D is given by

$$Info(D) = -\sum_{i=1}^{m} p_i \log_2(p_i),$$

- Let D, the data partition, be a training set of class-labeled tuples. Suppose the class label attribute has m distinct values defining m distinct classes, $C_i$ (here I = 1 to m); $p_i = s_i/s$; s= no. of samples; $s_i$ = no. of samples in class label $C_i$ ; *Info*(*D*) is also known as the **entropy** of *D*

# ID3--Continued

- suppose we were to partition the tuples in *D* on some attribute *A* having *v* distinct values, $[a_1, a_2, \ldots, a_v]$, as observed from the training data. If *A* is discrete-valued, these values correspond directly to the *v* outcomes of a test on *A*. Attribute *A* can be used to split *D* into *v* partitions or subsets, $[D_1, D_2, \ldots, D_v]$, where $D_j$ contains those tuples in *D* that have outcome $a_j$ of *A*

$$Info_A(D) = \sum_{j=1}^{v} \frac{|D_j|}{|D|} \times Info(D_j).$$

- Here, $|D_j| / |D|$ = acts as the weight of the *j*thpartition; $Info_A(D)$ is the expected information required to classify a tuple from *D* based on the partitioning by *A*.
- $Info(D_j) = -\sum_{i=1}^{m} p_{ij} \log_2(p_{ij})$; $p_{ij} = s_{ij} / |D_j|$ ; $s_{ij}$ = no. of samples belongs to class label $C_i$ and having the attribute value $a_j$

# ID3--Continued

- Information gain is defined as the difference between the original information requirement (i.e., based on just the proportion of classes) and the new requirement (i.e., obtained after partitioning on *A*).

$$Gain(A) = Info(D) - Info_A(D).$$

- In other words, *Gain*(*A*) tells us how much would be gained by branching on *A*. It is the expected reduction in the information requirement caused by knowing the value of *A*. The attribute *A* with the highest information gain, *Gain(A*), is chosen as the splitting attribute at node *N.*

# Problem statement: Find out Test Attribute

| *RID* | *age* | *income* | *student* | *credit_rating* | *Class: buys_computer* |
|---|---|---|---|---|---|
| 1 | youth | high | no | fair | no |
| 2 | youth | high | no | excellent | no |
| 3 | middle_aged | high | no | fair | yes |
| 4 | senior | medium | no | fair | yes |
| 5 | senior | low | yes | fair | yes |
| 6 | senior | low | yes | excellent | no |
| 7 | middle_aged | low | yes | excellent | yes |
| 8 | youth | medium | no | fair | no |
| 9 | youth | low | yes | fair | yes |
| 10 | senior | medium | yes | fair | yes |
| 11 | youth | medium | yes | excellent | yes |
| 12 | middle_aged | medium | no | excellent | yes |
| 13 | middle_aged | high | yes | fair | yes |
| 14 | senior | medium | no | excellent | no |

# Solution:

- Class P: *buys_computer* = "yes"
- Class N: *buys_computer* = "no"

$$Entropy(D) = -\frac{9}{14}\log_2(\frac{9}{14}) - \frac{5}{14}\log_2(\frac{5}{14}) = 0.940$$

- Compute the expected information requirement for each attribute: start with the attribute *age*

$$Gain(age, D)$$
$$= Entropy(D) - \sum_{v \in \{Youth, Middle-aged, Senior\}} \frac{|S_v|}{|S|} Entropy(S_v)$$
$$= Entropy(D) - \frac{5}{14} Entropy(S_{youth}) - \frac{4}{14} Entropy(S_{middle_aged}) - \frac{5}{14} Entropy(S_{senior})$$
$$= 0.246$$

$$Gain\ (income, D) = 0.029$$
$$Gain\ (student, D) = 0.151$$
$$Gain\ (credit_rating, D) = 0.048$$



$Entropy(S_{youth}) = -\sum_{i=1}^{2} p_{i1} \log_2(p_{i1})$

$= - p_{11}\log_2(p_{11}) - p_{21}\log_2(p_{21})$

$= -2/5 \log_2(2/5) - 3/5 \log_2(3/5)$

$= 0.971$

Here, $p_{11} = s_{11}/|D_1| = 2/5$

$p_{21} = s_{21}/|D_1| = 3/5$

log2 X = log10 X / log10 2

$Entropy(S_{middle}) = -\sum_{i=1}^{2} p_{i2} \log_2(p_{i2})$

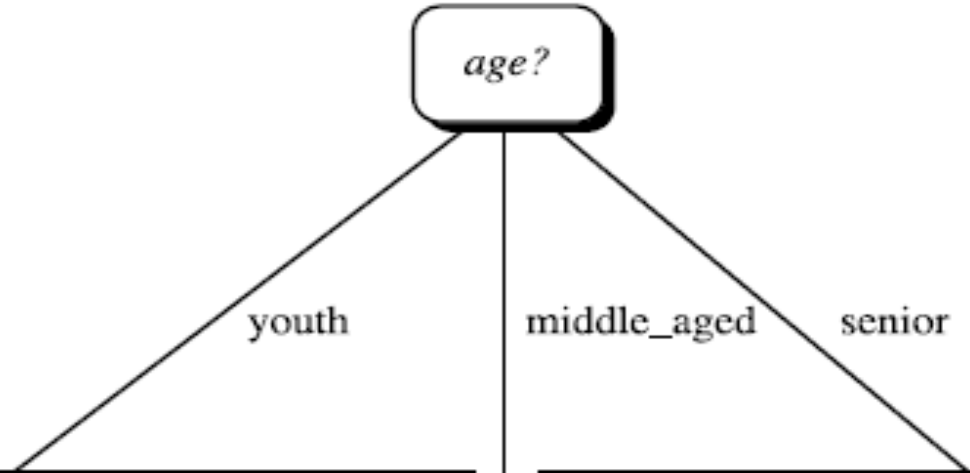
$= - p_{12}\log_2(p_{12}) - p_{22}\log_2(p_{22})$

$= -4/4 \log_2(4/4) - 0/4 \log_2(0/4)$

$= 0$

Here, $p_{12} = s_{12}/|D_2| = 4/4$

$p_{22} = s_{22}/|D_2| = 0/4$

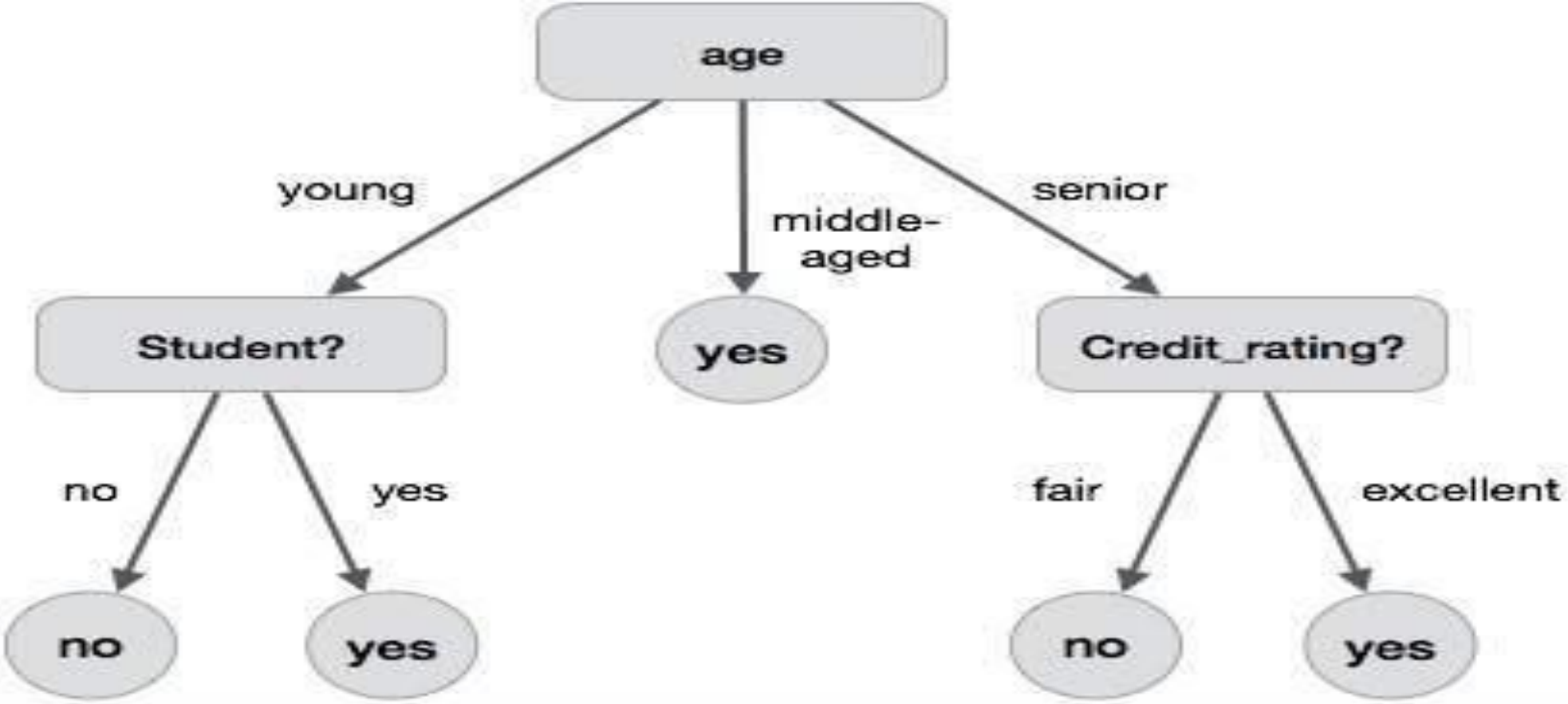



| *income* | *student* | *credit_rating* | *class* |
|---|---|---|---|
| high | no | fair | no |
| high | no | excellent | no |
| medium | no | fair | no |
| low | yes | fair | yes |
| medium | yes | excellent | yes |

| *income* | *student* | *credit_rating* | *class* |
|---|---|---|---|
| medium | no | fair | yes |
| low | yes | fair | yes |
| low | yes | excellent | no |
| medium | yes | fair | yes |
| medium | no | excellent | no |

| *income* | *student* | *credit_rating* | *class* |
|---|---|---|---|
| high | no | fair | yes |
| low | yes | excellent | yes |
| medium | no | excellent | yes |
| high | yes | fair | yes |

# Decision Tree




X = (age = youth, income = medium, student = yes, credit = fair)  Class label=?

# Extracting Rules from Decision Tree

| | | |
|---|---|---|
| *R*1: IF *age* = *youth* | AND *student* = *no* | THEN *buys_computer* = *no* |
| *R*2: IF *age* = *youth* | AND *student* = *yes* | THEN *buys_computer* = *yes* |
| *R*3: IF *age* = *middle_aged* | | THEN *buys_computer* = *yes* |
| *R*4: IF *age* = *senior* | AND *credit_rating* = *excellent* | THEN *buys_computer* = *no* |
| *R*5: IF *age* = *senior* | AND *credit_rating* = *fair* | THEN *buys_computer* = *yes* |

# Assignment:

| Outlook | Temperature | Humidity | Windy | Play |
|---|---|---|---|---|
| Sunny | Hot | High | False | ***No*** |
| Sunny | Hot | High | True | ***No*** |
| Overcast | Hot | High | False | ***Yes*** |
| Rainy | Mild | High | False | ***Yes*** |
| Rainy | Cool | Normal | False | ***Yes*** |
| Rainy | Cool | Normal | True | ***No*** |
| Overcast | Cool | Normal | True | ***Yes*** |
| Sunny | Mild | High | False | ***No*** |
| Sunny | Cool | Normal | False | ***Yes*** |
| Rainy | Mild | Normal | False | ***Yes*** |
| Sunny | Mild | Normal | True | ***Yes*** |
| Overcast | Mild | High | True | ***Yes*** |
| Overcast | Hot | Normal | False | ***Yes*** |
| Rainy | Mild | High | True | ***No*** |

# It will be Continued….